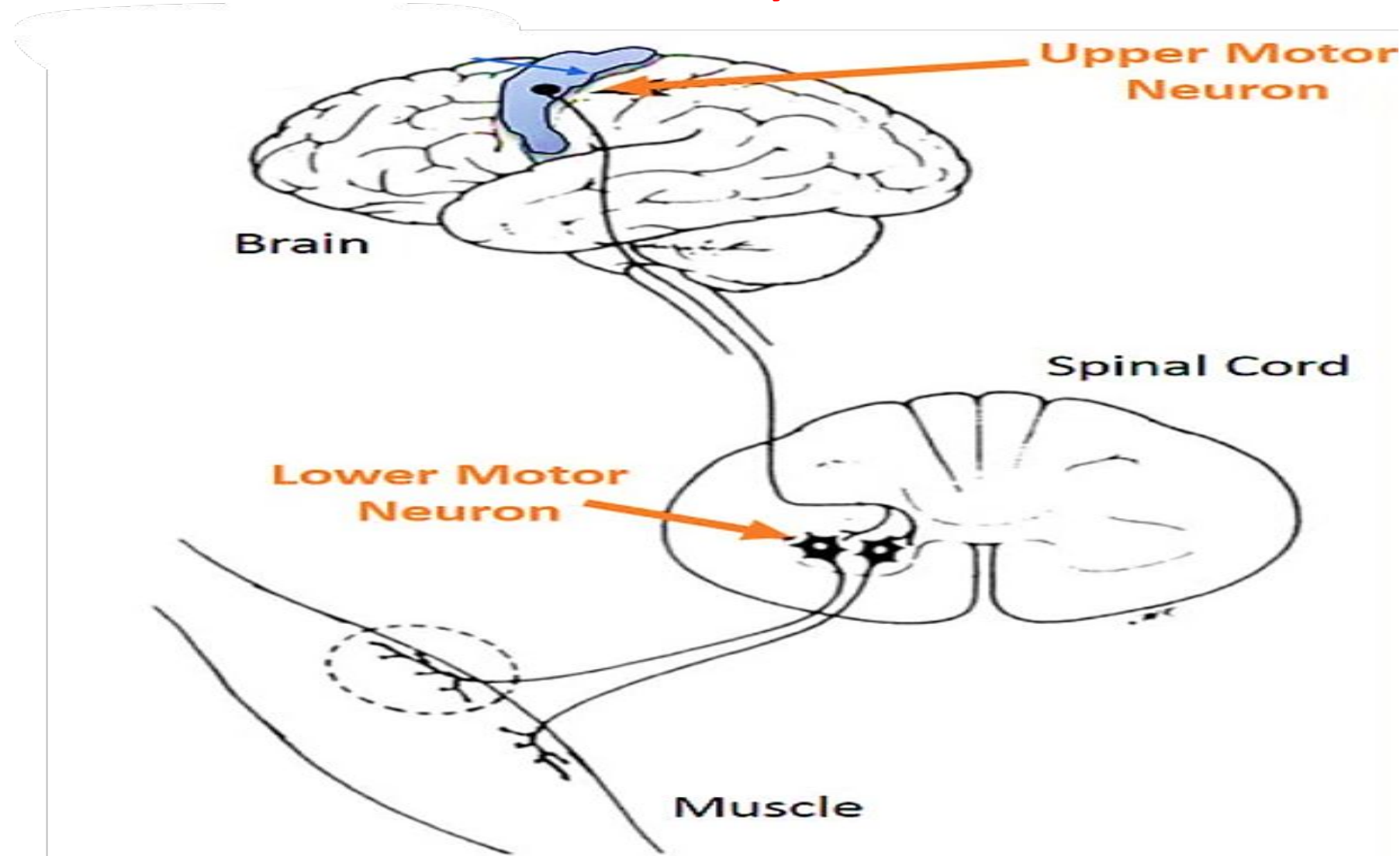


Differentiate UMN and LMN lesions, Identify Clinical features correlate lesions with signs

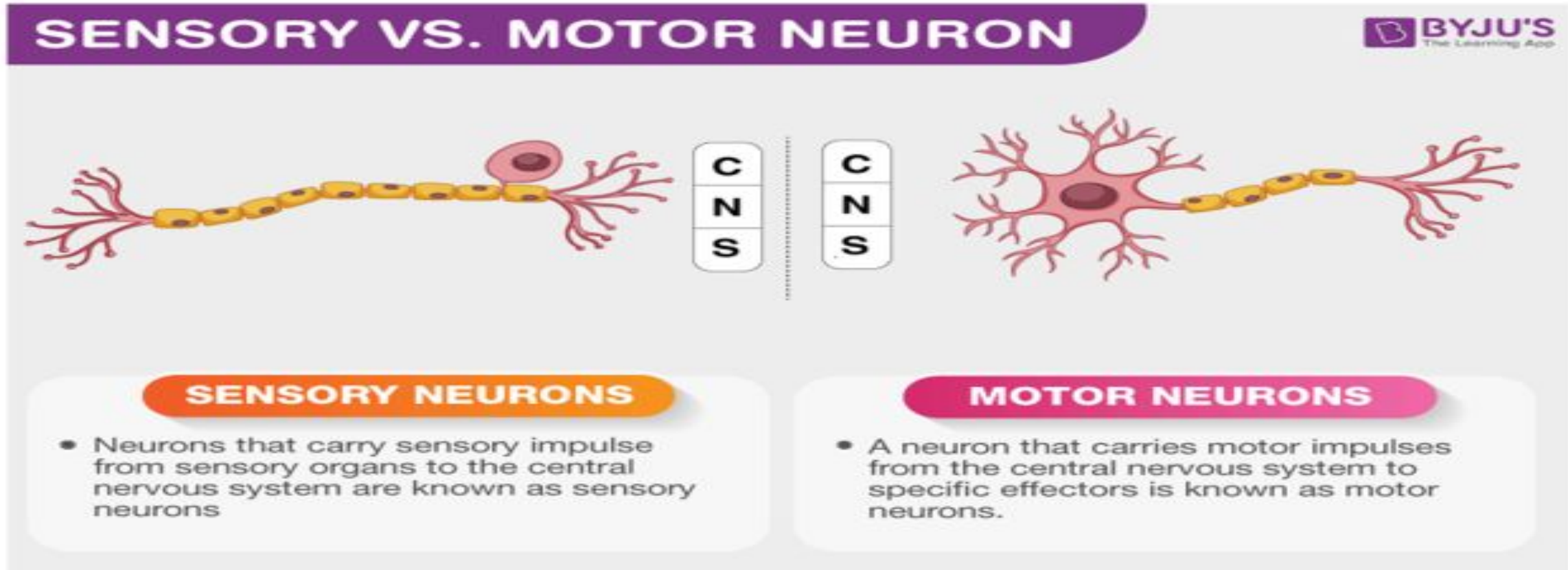


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What is Motor Neurons ?

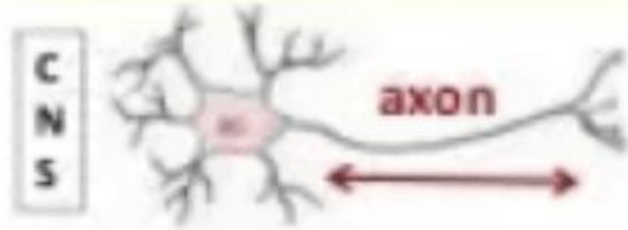
Motor neurons are specialized nerve cells in the central nervous system that transmit electrical signals from the brain and spinal cord to muscles and glands, triggering voluntary and involuntary actions like walking, breathing, and heart beating.

Neurons of the Nervous system are divided into:-



Sensory and Motor Neurons

Motor neurone



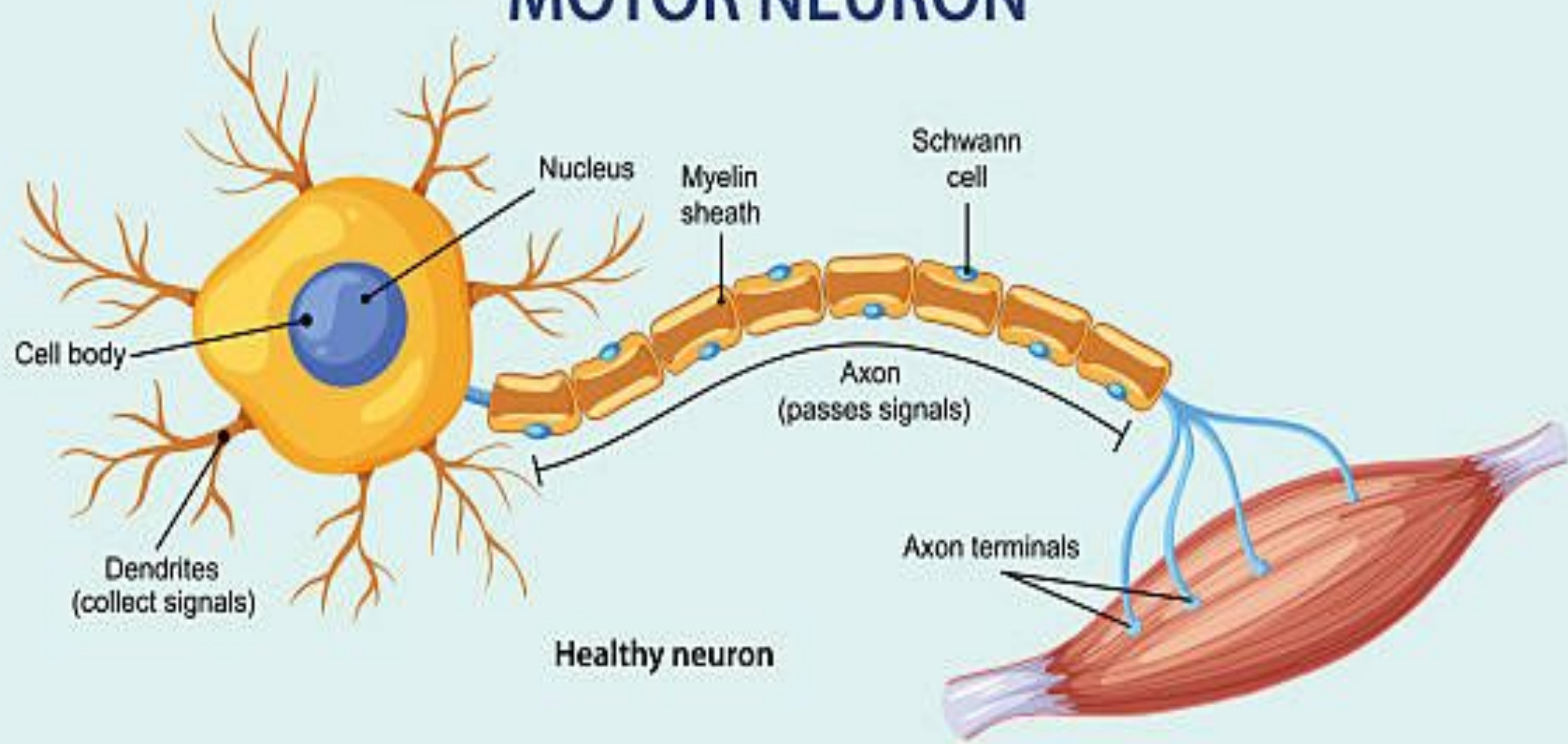
1. Long axon
2. No receptor
3. Cell body terminal & has dendrites
4. Many short dendrons

Sensory neurone



1. Short axon
2. Receptor
3. Cell body at the side; no dendrites
4. One long dendron

MOTOR NEURON



MOTOR NEURON:-

Neurons of the motor system are divided into upper motor neurons and lower motor neurons, depending upon their location and Functions

Upper Motor Neurons (UMNs):

Origin: Start in the brain ,Cerebral cortex or brainstem and send signals down to the spinal cord.

Function: Carry signals down to the spinal cord or brainstem to direct lower motor neurons.

Neurotransmitter: Glutamate.

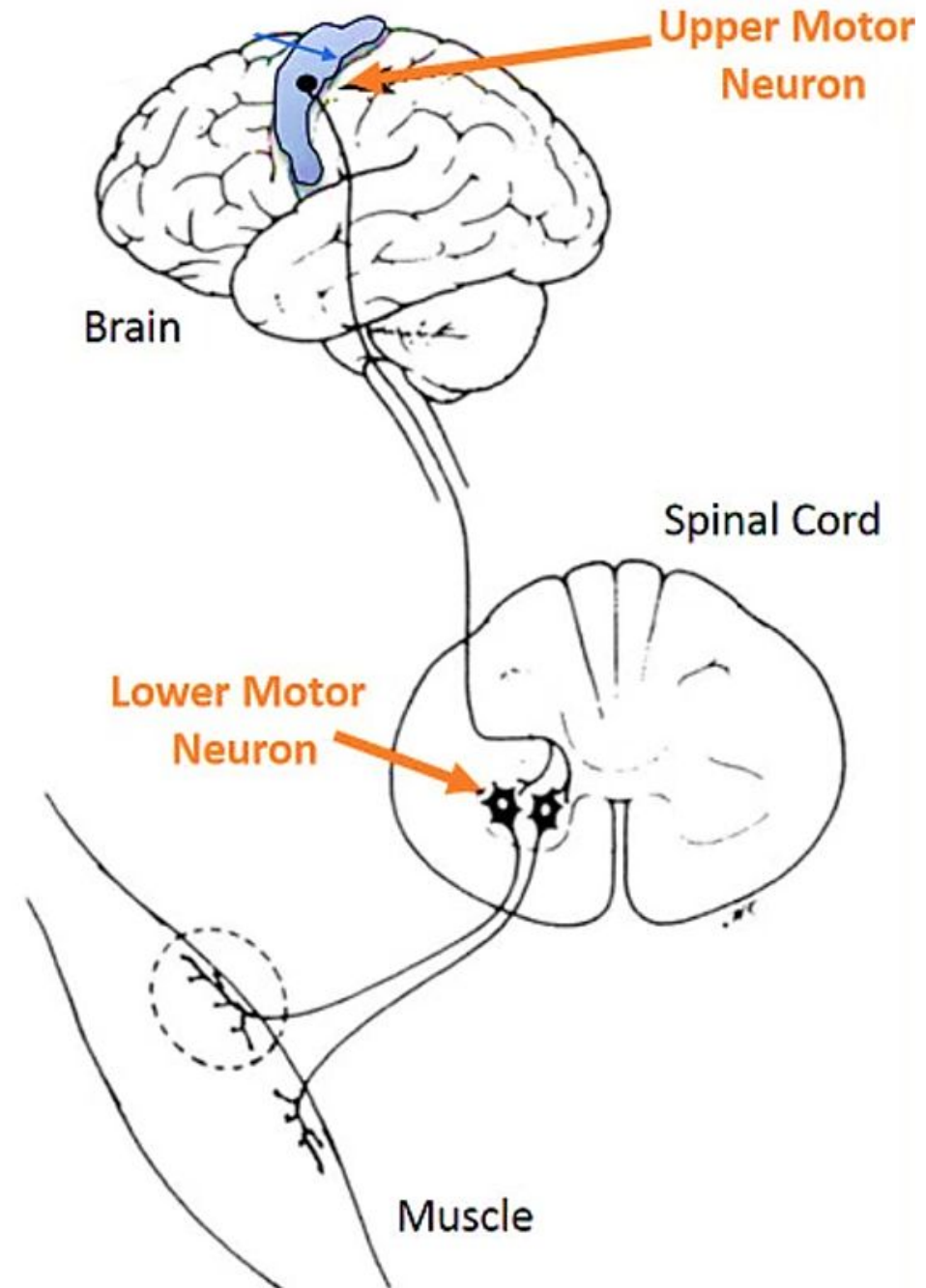
Lower Motor Neurons (LMNs):

Origin: Anterior horn of the spinal cord or cranial nerve nuclei in the brainstem.

Function: Carry signals from the spinal cord out to the body's muscles and glands.

Directly innervate effector muscles and peripheral organs.

Neurotransmitter: The primary neurotransmitter used by lower motor neurons is acetylcholine.



Upper Vs Lower Motor Neuron

Upper Motor Neuron	Lower Motor Neuron
It is entirely located in the central nervous system.	It is either located in the spinal cord grey matter or within the cranial nerve nuclei in the brain stem.
Transmits motor impulses from the brain to the synapses of the lower motor neuron.	Collects the transmitted motor impulses from the upper motor neuron and conducts them to the muscles in the body.
Cell bodies of upper motor neurons are bigger than lower motor neuron cell bodies.	Cell bodies are comparatively smaller than the upper motor neurons.
Classified on the basis of the pathways they are participating in.	Classified on the basis of the type of muscle fibre they innervate.
They form synapses with the lower motor neurons.	They form synapses with the muscles in the body.
Upper motor neuron lesions produce spastic hyperreflexia and muscle weakness.	Lower motor neuron lesions result in muscular atrophy, flaccid muscle weakness, fasciculation and hyporeflexia.

Motor Neuron Lesion

An upper motor neuron (UMN) lesion affects nerve pathways in the brain and spinal cord, causing stiff muscles, high reflexes, and a positive Babinski sign.

A lower motor neuron (LMN) lesion affects nerves from the spinal cord to the muscles, causing limp muscles, low reflexes, muscle shrinking, and twitches

Upper Motor Neuron Lesion

Causes

An upper motor neuron (UMN) lesion happens when there is damage to the nerve pathways in the brain or spinal cord that control voluntary movement.

Common causes include

- ☐ strokes
- ☐ multiple sclerosis,
- ☐ traumatic brain or spinal injuries,
- ☐ brain tumors, and
- ☐ amyotrophic lateral sclerosis (ALS)

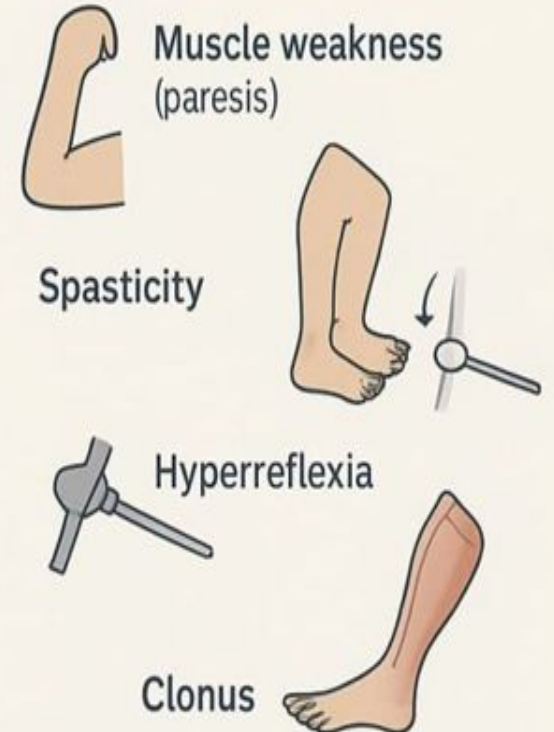
Upper Motor Neuron (UMN) Lesion

Damage to motor pathways in the brain or spinal cord before they synapse with lower motor neurons

Causes

- Stroke
- Traumatic brain injury or spinal cord injury
- Multiple sclerosis
- Brain tumors
- Cerebral palsy
- Neurodegenerative disorders as amyotrophic lateral sclerosis

Clinical Features



Upper motor neuron lesion signs:

An upper motor neuron (UMN) lesion will be in the central nervous system (brain and spinal cord).

Signs of Upper Motor Neuron Lesions

- ❑ **Hyperreflexia:** Exaggerated deep tendon reflexes (like knee or ankle jerks) because the brain no longer inhibits lower pathways.
- ❑ **Spasticity:** Increased, velocity-dependent muscle tone making muscles feel tight and rigid.
- ❑ **Positive Babinski Sign:** The big toe moves upward and the other toes fan out when you stroke the sole of the foot.
- ❑ **Clonus:** Rapid, rhythmic muscle contractions after a sudden stretch.
- ❑ **Weakness:** Loss of fine, voluntary motor control affecting specific muscle groups (often extensors in the arms and flexors in the legs).
- ❑ **Minimal Atrophy:** Only mild, disuse-related muscle wasting over a long period, unlike the severe wasting seen in lower motor neuron damage.
- ❑ **Absent Fasciculations:** Involuntary muscle twitches are generally not present

Lower Motor Neuron Lesion

A lower motor neuron (LMN) lesion damages the nerve pathways connecting the spinal cord or brainstem to the skeletal muscles, resulting in weak, limp muscles, low reflexes, and muscle shrinkage.

Lower motor neurons control movement in the arms, legs, chest, face, throat, and tongue.

Mixed upper and lower motor neuron diseases include multiple sclerosis.

Common causes include

- Trauma,
- Viral infections (Poliomyelitis)
- Genetic disorders (Spinal muscular atrophy (SMA)).
- Nerve compression
- Peripheral Nerve injury (Diabetic Neuropathy)

Signs and Symptoms

- **Flaccid paralysis:** Muscles become loose, soft, and weak.
- **Muscle atrophy:** Significant shrinking and wasting of the affected muscles.
- **Fasciculations:** Small, involuntary muscle twitches visible under the skin.
- **Hyporeflexia:** Decreased or totally absent deep tendon reflexes.
- **Decreased tone:** Low muscle tone, known as hypotonia.
- **Negative Babinski sign:** The big toe bends downward when the sole of the foot is stroked

Lower Motor Neuron Lesion findings:

A lower motor neuron (LMN) lesion affects anywhere from the anterior horn cell to the muscle.

On neurological examination, typical signs of a lower motor neuron lesion include:

- Variable patterns of weakness
- Marked atrophy or muscle wasting
- Fasciculations present in the associated muscle group
- Reduced tone
- Reduced or absent reflexes
- Babinski sign absent – downward going digits



A summary of upper motor neuron (UMN) vs lower motor neuron (LMN) lesions.

	Upper motor neurone	Lower motor neurone
Site of the lesion	Cerebral hemispheres, cerebellum, brainstem, spinal cord	Anterior horn cell, motor nerve roots, peripheral motor nerves
Inspection	No fasciculations or significant wasting	Wasting and fasciculation of muscles
Tone	Increased (spasticity or rigidity) +/- ankle clonus	Decreased (hypotonia, flaccid) or normal
Power	Reduced with pyramidal pattern of weakness (extensors weaker than flexors in arms, and vice versa in legs)	Reduced in distribution of affected motor root/nerve
Reflexes	Exaggerated or brisk (hyperreflexia)	Reduced or absent (hyporeflexia or areflexia)
Plantar reflex	Up going/extensor (Babinski positive)	Normal (down going/flexor) or no movement
Examples of conditions	• Ischemic or hemorrhagic stroke (including brainstem strokes) • Amyotrophic lateral sclerosis • Multiple sclerosis	• Peripheral nerve trauma/compression • Spinal muscular atrophy • Amyotrophic lateral sclerosis • Guillain-Barré syndrome • Poliomyelitis

Conditions in which Upper Motor Neuron (UMN) Lesions can apply:

- 1.Stroke:** This is one of the most common causes of UMN lesions. Strokes can cause brain damage that disrupts the normal functioning of the UMNs, leading to symptoms such as muscle weakness, spasticity, and hyperreflexia.
- 2.Multiple Sclerosis:** MS is a neurodegenerative disease that causes inflammation and scarring of the myelin sheath, which can disrupt the functioning of the UMNs.
- 3.Amyotrophic Lateral Sclerosis (ALS):** Also known as Lou Gehrig's disease, ALS causes degeneration of both UMNs and LMNs, leading to progressive muscle weakness and paralysis.
- 4.Cerebral Palsy:** This disorder is caused by abnormal development or damage to parts of the brain that control movement, balance, and posture, which often involves UMNs.
- 5.Brain Tumors:** Tumors in the brain can cause pressure or direct damage to the UMNs, leading to motor deficits.

Conditions in which Lower Motor Neuron (LMN) Lesions can apply:

- 1. Poliomyelitis:** This viral infection primarily affects LMNs, leading to muscle weakness and paralysis.
- 2. Peripheral Neuropathy:** This is a general term for disorders of the peripheral nerves, which can include LMNs.
- 3. Guillain-Barre Syndrome:** This is an autoimmune disorder that causes inflammation of the nerves, which can affect the LMNs.
- 4. Bell's Palsy:** This causes sudden weakness in the muscles of one half of the face, often due to damage to the facial nerve.
- 5. Amyotrophic Lateral Sclerosis (ALS):** ALS affects both UMNs and LMNs.
- 6. Spinal Muscular Atrophy:** This is a genetic disorder that causes wasting of the skeletal muscles due to LMN

Identify Clinical features correlate lesions with signs

UMN vs LMN lesion

Feature	Upper motor neuron lesion	Lower motor neuron lesion
Paralysis	Groups of muscles of one or more limbs paralyzed	Individual muscles paralyzed
Muscle tone	Spasticity/rigidity (spastic paralysis)	Flaccid (flaccid paralysis)
Deep tendon reflexes	Exaggerated	Absent
Superficial reflexes (abdominal & cremasteric)	Absent	Absent
Plantar response	Extensor (Babinski sign positive)	No response
Muscle atrophy	May not be marked (may occur later)	Will be early, severe & due to denervation
Fasciculations & fibrillations	Do not occur	Are common

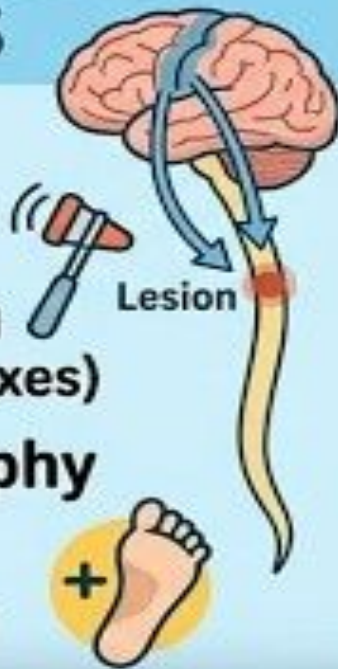
MOTOR NEURON SIGNS COMPARISON

UPPER VS. LOWER

UPPER MOTOR NEURON (UMN) SIGNS



- **Hypertonia**
(Spasticity)
- **Hyperreflexia**
(Increased Reflexes)
- **Minimal Atrophy**
- **Positive Babinski Sign**



LOWER MOTOR NEURON (LMN) SIGNS



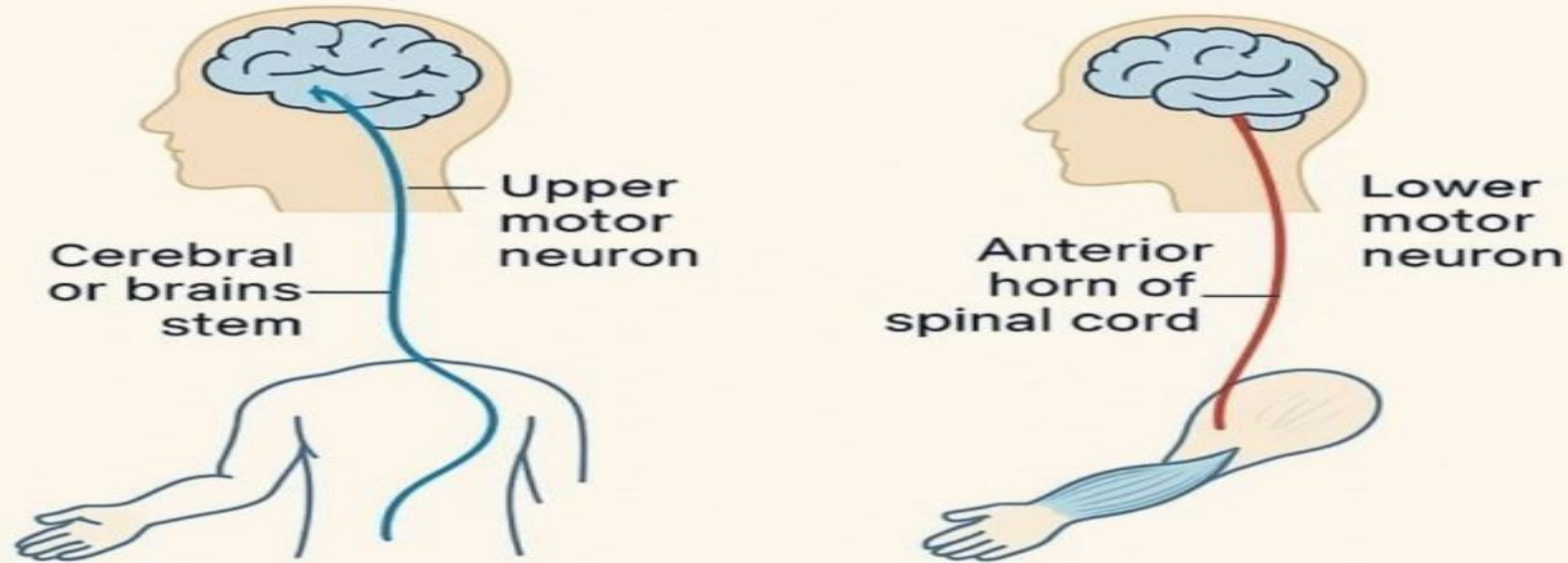
- **Hypotonia**
(Flaccidity)
- **Hyporeflexia**
(Decreased Reflexes)
- **Marked Muscle Atrophy**
- **Fasciculations**
(Twitches)



WATCH NOW!



Upper vs Lower Motor Neurons



Clinical Differences: UMN vs LMN Lesions

Feature	Upper Motor Neuron Lesion	Lower Motor Neuron Lesion
Muscle tone	Increased (spasticity)	Decreased (flaccidity)
Reflexes	Hyperreflexia	Hyporeflexia or areflexia
Babinski sign	Positive (toes fan upward)	Negative
Muscle atrophy	Minimal or disuse atrophy	Severe, neurogenic atrophy
Fasciculations	Absent	Present
Paralysis	Spastic paralysis	Flaccid paralysis

Feature	Upper Motor Neuron Lesion (UMNL)	Lower Motor Neuron Lesion (LMNL)
Muscle Tone	Increased (spasticity)	Decreased (flaccidity)
Muscle Atrophy	Mild atrophy (due to disuse over time)	Significant atrophy (due to denervation)
Muscle Strength	Weakness (more pronounced in specific muscle groups)	Weakness (more generalized, affecting individual muscles)
Distribution of Weakness	Contralateral weakness (opposite side of lesion)	Ipsilateral weakness (same side as lesion)
Reflexes	Hyperreflexia (exaggerated reflexes)	Hyporeflexia or areflexia (diminished or absent reflexes)
Fasciculations	Absent	Present
Clonus	Often present	Absent
Babinski Reflex	Positive (upward response)	Negative (downward response)
Common etiologies	-Stroke (Cerebrovascular accident) -Traumatic Brain Injury -Cerebral Palsy -Multiple Sclerosis -Spinal Cord Injury -Amyotrophic Lateral Sclerosis (in later stages)	-Peripheral Neuropathy -Guillain-Barré Syndrome -Spinal Muscular Atrophy -Poliomyelitis -Radiculopathy (nerve root compression) -Amyotrophic Lateral Sclerosis (early stages)

UMN vs LMN



UPPER MOTOR NEURON (UMN)

VS

LOWER MOTOR NEURON (LMN)

SIGNS:



Increased Reflexes



Spastic Paralysis



Babinski Sign



Clasp-Knife Rigidity



CAUSES:

- Stroke
- Multiple Sclerosis

SIGNS:



Decreased Reflexes

Decreased Reflexes



Flaccid Paralysis



Atrophy



Fasciculations



CAUSES:

- Peripheral neuropathy
- Amyotrophic Lateral Sclerosis

Key Differences Between UMN and LMN Lesions

Clinical Feature	UMN Lesion	LMN Lesion
Location of Damage	Cortex, brainstem, corticospinal tract, or spinal cord above anterior horn	Anterior horn cells, nerve roots, peripheral nerves, cranial motor nuclei
Muscle Weakness	Present	Present
Muscle Tone	Increased (spasticity)	Decreased (flaccidity)
Deep Tendon Reflexes	Increased (hyperreflexia)	Decreased or absent (hyporeflexia/areflexia)
Pathological Reflexes	Present (e.g., positive Babinski sign)	Absent
Muscle Atrophy	Mild (due to disuse)	Marked (due to denervation)
Fasciculations	Absent	May be present
Pattern of Weakness	Typically affects groups of muscles, more generalized	Focal or segmental, matching the distribution of affected nerve or root

Upper vs Lower Motor Neuron Lesion

Feature	Upper Motor Neuron (UMN)	Lower Motor Neuron (LMN)
Site of lesion	Cortex, internal capsule, corticospinal tract above anterior horn cell	Anterior horn cell, peripheral nerve, neuromuscular junction
Tone	↑ Increased (spasticity)	↓ Decreased (flaccidity)
Reflexes	Hyperreflexia (exaggerated)	Hyporeflexia / Areflexia
Plantar reflex	Babinski positive (upgoing)	Normal or absent
Weakness pattern	Patterned: Extensors weaker in upper limb, Flexors weaker in lower limb	Localized: Depends on affected nerve/muscle (no fixed flexor/extensor rule)
Wasting	Mild / disuse atrophy	Marked / severe atrophy
Fasciculations	Absent	Present